

Learned Trust Amplifies Spurious Features into Group Structure

[TODO: author block]

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Abstract

Agents that learn from other agents must estimate how far to trust each source. We study a population of N neural-network learners under a Bayesian online rule that jointly updates each agent’s weights and an explicit, dynamical reliability estimate for every other agent. Into this population we introduce a *representation error*: a fraction f_d of agents extend their encoding of each message with a feature of strength d that carries no information about the task and depends only on the sending agent’s group label. Individually this is a mild misspecification, and against a randomly assigned label it does nothing but degrade learning. Against the group label it does not stay mild: above a sharp boundary in (d, f_d) the population splits into mutually distrustful groups aligned with the label. We map the full phase diagram and identify two distinct discriminatory regimes, one in which the split still tracks learned representations and one in which trust tracks the label almost independently of them. Ablations isolate the cause. Freezing the reliability sector, or removing the sign reversal by which distrusted sources are anti-learned, eliminates the transition; Hebbian and Perceptron updates under the same perturbation do not produce it at all. Finally, task diversity $\alpha = P/K$ controls the *order* in which structure appears: with few tasks, trust structure forms first and drags representations behind it, and with many the reverse. The amplification is therefore a property of reliability-weighted learning rather than of the architecture, and it is strongest in exactly the regime such weighting is usually adopted for.

1 Introduction

A learner that takes supervision from other learners cannot treat all of its sources alike. Some are better than others, some are adversarial, and the standard response is to estimate a per-source reliability and weight incoming gradients by it. This is one of the most widely deployed ideas in machine learning: it is the Dawid–Skene construction in crowdsourcing, annotator modelling in preference learning, client reweighting in federated optimisation, and peer weighting in decentralised training. In each case reliability estimation is introduced defensively — it is the component that makes learning *robust* to bad sources.

This paper reports a cost of that component that we have not seen stated. A per-source reliability estimate is a variable indexed by the source. If anything about how a learner encodes its inputs depends on *which* source sent them, the reliability estimate is the natural place for that dependence to accumulate. When the source-dependent part of the encoding happens to correlate with a group label, reliability estimation converts a local encoding defect into a global, self-reinforcing partition of the population — and does so with a sharp threshold rather than gracefully.

We make this precise in a population of N agents that exchange opinions about a fixed agenda of P issues. Each agent carries a classifier (the *opinion* sector) and, for every other agent, a scalar reliability estimate (the *trust* sector). Both are updated by the same rule, a Gaussian variational approximation to the Bayesian posterior that is optimal for a single teacher–student pair. The perturbation we inject is a representation error in the sense familiar from shortcut learning: a discriminating agent appends to each message a feature that is uninformative about the issue and determined entirely by the sender’s group label. It shifts one field by a constant d . Nothing in the model expresses a preference for either group, no training data is biased, and the label is uncorrelated with every issue.

Contributions.

- We identify **learned per-source reliability weighting as an amplifier of spurious features**. A perturbation that is harmless when keyed to a random per-source bit produces a population-wide split when keyed to a group label of the same magnitude (§7, condition A0).
- We map the **phase diagram** in the strength d of the spurious feature and the fraction f_d of agents carrying it, extract the boundary $d_c(f_d)$, and separate **two distinct discriminatory regimes**: one in which the split still tracks learned representations, and one in which trust tracks the group label almost independently of them (§6).
- We **isolate the mechanism by ablation**. Freezing the trust sector, or removing the sign reversal by which distrusted sources are anti-learned, removes the transition; Hebbian and Perceptron updates under the identical perturbation do not exhibit it (§7).
- We show that **task diversity $\alpha = P/K$ determines the order in which structure crystallises** — trust before representation, or the reverse — which is a statement about what a population of learners aligns first, and is independent of whether any spurious feature is present at all (§8).

The frame. The update rule we study is optimal in a well-defined sense, for a pair. Deployed in a population it is pathological, and the pathology is not a bug in the derivation — it follows from the two features that make the rule good at its stated job. We read this as objective misspecification by scale: an objective correct at the scale it was derived for, wrong at the scale it is used at. We state this once here and once in §10, and otherwise let the ablations carry the argument.

[TODO: one short paragraph on why this is not merely a restatement of model collapse — there the degradation is in the data distribution, here the data distribution is fixed and the pathology is in the weighting.]

2 Setup: a population of learners with per-source reliability

Agents. A population of N agents. Agent I holds an *opinion sector* — weights $\mathbf{w}_I \in \mathbb{R}^K$ with covariance C_I — and a *trust sector*: for every other agent J , a scalar $\mu_{J|I}$ with variance $V_{J|I}$. The belief state is the product of Gaussians

$$Q(\mathbf{w}_I, \mathbf{z}_I \mid \lambda_I) = \mathcal{N}(\hat{\mathbf{w}}_I, C_I) \prod_{J \neq I} \mathcal{N}(\mu_{J|I}, V_{J|I}), \quad (1)$$

where the means say what the agent holds and the variances say how firmly. Each agent also carries an immutable label $\kappa_I = \pm 1$, visible to every other agent and carrying no information about any issue.

Read $\mu_{J|I}$ as I 's estimate of the probability that J transmits the wrong label: it is a per-source reliability parameter, learned online, with an explicit uncertainty attached. Positive μ is distrust.

The covariances are also the learning rates. It is worth being explicit about the double role C_I and $V_{J|I}$ play, because it is what makes the dynamics of §3 interpretable. As posterior widths they are epistemic: they say how uncertain the agent is. But they also premultiply every update the agent ever makes — the weight step is proportional to C_r and the trust step to $V_{e|r}$ (Eq. 5) — so they are simultaneously the step sizes.

The degenerate limit makes the point. As $C_I \rightarrow 0$ the agent is *completely certain* of its opinion, and its weights stop moving: no message, from any source, on any issue, changes what it believes. As $V_{J|I} \rightarrow 0$ it is completely certain of how far to trust J , and that estimate stops moving whatever J subsequently says. With both at zero the agent is frozen — not because it has stopped receiving information, but because a Bayesian update to a delta-function prior returns the prior. Certainty and immobility are the same statement here, and no separate learning-rate hyperparameter appears anywhere in the model.

This is not a corner case to be excluded, because the dynamics drives the system toward it: both covariances shrink monotonically under the update (§3), so a population anneals rather than equilibrating, and *when* it is measured is a parameter of the experiment (Appendix D). It is also the reason the initialisation

$C_0 = I$, $V_0 = 1$ matters more than it looks: it sets the total amount of movement available to the population over its whole history.

Tasks. An issue is a unit vector $\hat{\mathbf{x}} \in \mathbb{R}^K$; the agenda is a fixed set of P of them. Agent I 's opinion on an issue is $\sigma_I = \text{sign}(\mathbf{w}_I \cdot \hat{\mathbf{x}})$. Because each agent is a classifier rather than a scalar state, two agents can agree on some issues and disagree on others — the property that motivates using a network here at all, and that a two-state model cannot represent. We write $\alpha = P/K$ for the **task diversity** of the agenda, and it turns out to control more than the difficulty of the learning problem (§8).

One interaction. Draw an issue $\hat{\mathbf{x}}$, an emitter e and a receiver r . The emitter transmits its opinion $\sigma_e = \pm 1$. The receiver updates both sectors. Two independent corruptions stand between the emitter's computation and the receiver's update:

1. **Channel noise.** The transmitted σ_e differs from the true label with probability ε . Modelling $\varepsilon = \Phi(z)$ with $z \sim \mathcal{N}(\mu_{e|r}, V_{e|r})$ keeps the update closed in the Gaussian family; the trust sector *is* the receiver's posterior over the channel's flip probability.
2. **Encoding mismatch.** The vector the receiver parses is not the one the emitter parsed. They are jointly Gaussian with $\mathbb{E}[x_i y_i] = v_p$, and the choice $v_p = \|\mathbf{w}\|^{-2}$ makes an inexperienced agent (small weight norm) a worse parser. Details in Appendix A.

The second is where the perturbation of §4 enters. It is a statement about *representation*, not about preference: the two agents do not encode the same message into the same vector.

[TODO: one sentence placing this against decentralised SGD / federated setups — same structure, but there the peer weights are usually a hyperparameter or a robust-aggregation rule rather than a learned posterior.]

3 The learning rule

Marginalising both corruptions of §2 gives the likelihood of the received opinion, and projecting the resulting posterior back onto the Gaussian family of Eq. 1 by maximum entropy gives a closed-form online update. The derivation is standard and deferred to Appendix A; what matters in the main text is the result and two of its properties.

Two fields. Everything the update depends on enters through two scalars, each a raw field divided by an *internal response uncertainty*:

$$h_w = \frac{(\hat{\mathbf{w}}_r \cdot \hat{\mathbf{x}}) \sigma_e}{\gamma_C}, \quad h_\mu = \frac{\mu_{e|r}}{\gamma_V}, \quad \gamma_C = \sqrt{1 + \hat{\mathbf{x}} \cdot C_r \hat{\mathbf{x}}}, \quad \gamma_V = \sqrt{1 + V_{e|r}}. \quad (2)$$

$h_w > 0$ means the receiver already agrees with the message; $h_\mu > 0$ means it distrusts the source.

The evidence is an exclusive-or. The marginal likelihood reduces to

$$Z = \Phi(h_w) + \Phi(h_\mu) - 2\Phi(h_w)\Phi(h_\mu), \quad (3)$$

which is $P(a) + P(b) - 2P(a)P(b)$: the probability that exactly one of the two holds. Z is large in the two *consonant* configurations — agree with a trusted source, disagree with a distrusted one — and collapses to zero in the two *dissonant* ones. Surprise, in this model, is not an assumption; it is what the marginal likelihood does.

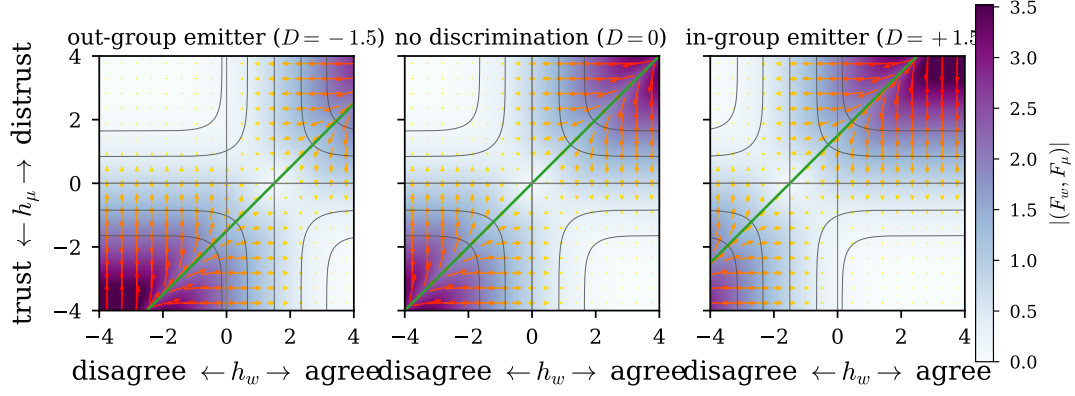


Figure 1: The flow induced on (h_w, h_μ) by one interaction, at $D = -d$ (left), $D = 0$ (centre) and $D = +d$ (right). Learning moves the receiver out of the dissonant quadrants. At $D = 0$ the two consonant basins are symmetric, so the population polarises but the split is uncorrelated with the label; a nonzero D tilts the basin boundary, which is the entire mechanism of §4.

The update. The four modulation functions are the log-derivatives of Eq. 3,

$$F_w = (1 - 2\Phi(h_\mu)) \frac{g(h_w)}{Z}, \quad F_\mu = (1 - 2\Phi(h_w)) \frac{g(h_\mu)}{Z}, \quad F_C = -F_w(F_w + h_w), \quad F_V = -F_\mu(F_\mu + h_\mu), \quad (4)$$

with g the standard normal density, and the receiver moves

$$\begin{aligned} \hat{\mathbf{w}}_r &+= \frac{F_w}{\gamma_C} \sigma_e C_r \hat{\mathbf{x}}, & C_r &+= \frac{F_C}{\gamma_C^2} (C_r \hat{\mathbf{x}})(C_r \hat{\mathbf{x}})^\top, \\ \mu_{e|r} &+= \frac{F_\mu}{\gamma_V} V_{e|r}, & V_{e|r} &+= \frac{F_V}{\gamma_V^2} V_{e|r}^2. \end{aligned} \quad (5)$$

The opinion update is Hebbian in $\sigma_e \hat{\mathbf{x}}$, with a tensorial, adaptive, F_w -modulated learning rate. Note the symmetry $F_w(x, y) = F_\mu(y, x)$ and $F_C(x, y) = F_V(y, x)$: the two sectors run the same dynamics with their arguments exchanged. This is what the scaling by γ_C, γ_V buys, and it is why group structure can appear in either sector first (§8).

3.1 Two properties that do all the work

Anti-learning. The prefactor $(1 - 2\Phi(h_\mu))$ in F_w is *negative* whenever the source is distrusted. The receiver then moves its weights away from what the source transmitted — it learns the opposite. Effective antiferromagnetic couplings appear in a system where nothing antiferromagnetic was put in, and this is the engine of every collective effect we report. Ablation A2 in §7 removes exactly this sign and nothing else.

Blame attribution. On a surprise, one sector must yield, and Eq. 4 decides which: the sector whose field is *smaller in magnitude* absorbs the change, with the crossover at $|h_\mu| = |h_w|$. A receiver with a firm opinion and a weak read on its source revises the source’s reliability; one with a weak opinion and a firm read on its source revises its opinion. Neither behaviour was specified. Both follow from Eq. 3.

Annealing. F_C and F_V are negative almost everywhere, so both covariances shrink monotonically. Since the covariances are also the step sizes (§2), the population does not approach a stationary state so much as run out of the ability to move: it is annealing toward the frozen $C, V \rightarrow 0$ limit, and every agent becomes progressively harder to change as it becomes more certain. Two consequences we take seriously rather than treating as nuisances. *When* the observables are measured is a parameter of the experiment, and we report it (Appendix D) where the source draft leaves it unspecified. And the *order* in which the two sectors organise is a real feature of the trajectory rather than an artefact of stopping early, which is what §8 measures.

[TODO: decide whether the modulation-surface figure (modulation_surfaces, modulation_contours) earns main-text space or moves wholesale to Appendix A. Current recommendation: appendix. The two properties above are the content, and Fig. 1 already shows them acting.]

4 A spurious feature as a field shift

We now perturb the encoding, and only the encoding.

A fraction f_d of agents, drawn independently of their own label, extend their representation of an incoming message with a feature that carries no information about the issue and is determined entirely by the *sender's* label. This is the shortcut-learning setup moved from one classifier to a population: a feature that is predictive of nothing the task cares about, but perfectly predictive of a group membership. Because it enters the receiver's encoding of the message, its effect on Eq. 2 is an additive shift of the opinion field,

$$h_w \longrightarrow h_w + D[\kappa_r, \kappa_e], \quad (6)$$

with D a 2×2 matrix indexed by the receiver's and the emitter's labels. The main text uses the fully symmetric case, in which both groups are affected alike:

$$D = d \cdot \begin{pmatrix} +1 & -1 \\ -1 & +1 \end{pmatrix}, \quad d > 0. \quad (7)$$

The five asymmetric variants — one group only, one direction only, and so on — are implemented and reported in Appendix C.

Sign convention. With Eq. 7 and $d > 0$, a same-label sender's message arrives with h_w shifted *up*, toward agreement, and an opposite-label sender's shifted *down*, toward disagreement. Since F_μ carries the prefactor $(1 - 2\Phi(h_w))$, agreement builds trust and disagreement erodes it, so $d > 0$ is **in-group tolerance and out-group intolerance** and $d < 0$ is its reverse. We flag this explicitly because the sign is easy to get backwards, and because it fixes the orientation of the x axis in every figure that follows.

What has and has not been assumed. No training data is biased. No agent holds a preference over groups. The label is uncorrelated with every issue on the agenda, and f_d is independent of label, so neither group is over-represented among the perturbed agents. A discriminating agent runs *the same update* as every other agent; it has merely encoded the message badly, in a way that happens to depend on who sent it. Everything in §6 is downstream of Eq. 6 alone.

Why this should be harmless, and is not. An uninformative feature appended to an input is, for a single learner, just noise: it costs sample efficiency and nothing else. That intuition is correct here too, and we verify it — keying the same shift to a per-source *random* bit rather than the label degrades learning and produces no group structure (§7, A0). The difference is that the trust sector is indexed by source. A shift that is constant across a source's messages accumulates in $\mu_{e|r}$ rather than averaging out, and once μ is class-correlated the anti-learning prefactor of §3 makes the opinion sector class-correlated too. The feedback runs in both directions, which is what produces a threshold rather than a gradual degradation.

5 Observables

Two per-pair quantities, both signed and both bounded, and both independent of K and of the architecture — deliberately, so that the same measurements apply to any population of agents that can be asked what it thinks and whom it trusts.

- **Opinion alignment** $\rho_{IJ} = \cos(\mathbf{w}_I, \mathbf{w}_J) \in [-1, 1]$.
- **Trust** $\eta_{e|r} = 1 - 2\Phi(h_\mu) \in [-1, 1]$, equal to $+1$ for a fully trusted source and -1 for a fully distrusted one. Note that η is directional: $\eta_{I|J} \neq \eta_{J|I}$ in general.

With $G_{IJ} = \kappa_I \kappa_J$ the signed label agreement, the three pair correlations are, averaged over unordered pairs and each normalised to $[-1, 1]$,

$$R_{w\mu} = \langle (\eta_{I|J} + \eta_{J|I}) \rho_{IJ} \rangle, \quad R_{\mu c} = \langle G_{IJ} (\eta_{I|J} + \eta_{J|I}) \rangle, \quad R_{cw} = \langle G_{IJ} \rho_{IJ} \rangle. \quad (8)$$

$R_{\mu c}$ — how strongly trust tracks the group label — is the **discrimination order parameter**, and the one the primary claim is about. R_{cw} says whether the split is also visible in what agents believe, and separating the two is what distinguishes the regimes of §6.

Balance. Correlations do not distinguish an ordered population from a disordered one. Over triples,

$$B_\rho = \langle \rho_{IJ} \rho_{JK} \rho_{KI} \rangle, \quad B_\eta = \left\langle \frac{1}{2} (\eta_{IJ} \eta_{JK} \eta_{KI} + \eta_{JI} \eta_{IK} \eta_{KJ}) \right\rangle, \quad (9)$$

the second symmetrised because η is directional. Two coherent, mutually opposed blocs give $B = 1$ whatever they are made of; a random population gives 0; a population that cannot settle goes negative. B_ρ measures balance in the opinion sector and B_η in the trust sector, and reading them as a trajectory in the (B_ρ, B_η) plane is what §8 does.

Both aggregates have closed forms — for any M with unit diagonal, $\sum M_{IJ} M_{JK} M_{KI} = \text{tr}(M^3) - 3(\sum_{IJ} M_{IJ} M_{JI} - N) - N$ over distinct ordered triples — which replaces an $O(N^3)$ enumeration by two matrix products.

Naming. The source draft calls these B_I (“ideological”) and B_A (“affective”). We name them after the quantity each is built from, and we do not use “ideological” or “affective” anywhere: the w sector holds a classifier and the μ sector holds a reliability estimate. **[TODO: axis labels in frustration_maps and agenda_trajectories still read B_I / B_A and need regenerating as B_ρ / B_η .]**

6 The population splits above a sharp boundary

Figure 2 is the main result. Sweeping the strength d of the spurious feature and the fraction f_d of agents carrying it produces four regimes, and the two that matter are separated by a boundary that is sharp rather than gradual.

Polarisation is not the phenomenon. At $d = 0$ the population still splits — the anti-learning prefactor of §3 guarantees it — but the split is uncorrelated with the label ($R_{\mu c} \approx 0$, region II). This is worth stating early because it establishes what the spurious feature actually does: it does not *create* group structure, it *aligns* pre-existing group structure with a label that carries no information. The two consonant basins of Fig. 1 are symmetric at $d = 0$ and the population falls into one or the other arbitrarily; a nonzero d tilts them.

Two discriminatory regimes. Region (III) and region (IV) both have high $R_{\mu c}$ and would be reported identically by any single-number measure of group bias. They are structurally different. In (III), R_{cw} and $R_{w\mu}$ are also high: the population is split along the label *and* along what it believes, and the two are consistent. In (IV) the spurious feature is strong enough that $R_{\mu c}$ saturates while R_{cw} falls away — trust tracks the label almost independently of opinion, and an agent distrusts an out-group source regardless of whether it agrees with the content. The transition between them is not visible in $R_{\mu c}$ alone, which is an argument for reporting the three correlations separately.

Frustration. Regions (I) and (III) differ in a way that the correlations alone do not capture. Figure 4 shows B_ρ and B_η over the plane: $d > 0$ produces a split that is *balanced* — two internally coherent, mutually opposed blocs, low frustration — while $d < 0$ produces a population that cannot settle at all, with the balance measures driven negative and no stable partition. Over-tolerance toward the out-group is not the benign mirror image of intolerance; it is a different failure.

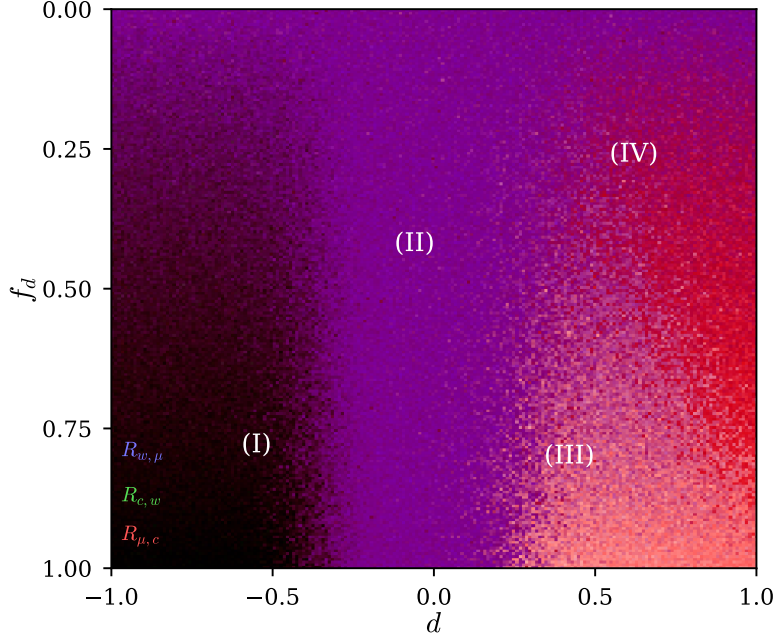


Figure 2: The (d, f_d) plane, with $R_{w\mu}$, $R_{\mu c}$ and R_{cw} composited into one map. **(I)** $d < 0$: over-tolerance to the out-group, heavily frustrated, no label-correlated order. **(II)** neutral — the population polarises, but the split is uncorrelated with the label. **(III)** discriminatory, with the split still tracking opinion. **(IV)** discriminatory, with trust tracking the label almost independently of opinion. The boundary between (II) and (III) is sharp in d and weakly dependent on f_d .

6.1 From a picture to a prediction

[NEEDS EXPERIMENT: extract $d_c(f_d)$ at a fixed $R_{\mu c}$ threshold, with error bars over seeds; fit it; repeat at $N \in \{20, 40, 80\}$ and attempt a finite-size collapse.]

As it stands Fig. 2 is a colour boundary in a pixel map, which supports the claim qualitatively and not more. The boundary should be extracted as a curve, given error bars over seeds, and checked for finite-size scaling. Two questions a reviewer will ask and we should answer before they do: does d_c depend on f_d as $1/f_d$, i.e. is only the total injected perturbation $f_d \cdot d$ what matters, or is there a genuine two-dimensional structure? And does the transition sharpen with N , as a genuine collective transition should?

[TODO: a single additional panel with a non-perceptron architecture, or with issues drawn from real sentence embeddings rather than Gaussian $\hat{x}$, removes the largest single objection to this section. It does not need to be the main result; it needs to exist.]

7 What causes it: four ablations

[NEEDS EXPERIMENT: this entire section. Nothing in it has been run. It is the section that decides whether the paper is a mechanism paper or a simulation study, and it should be built before anything else on the to-do list. Full specification in Appendix B.]

Section 6 establishes that the split happens. It does not establish what causes it. Four conditions, each swept over the same (d, f_d) grid with everything else held fixed, isolate the mechanism.

A0 — the spurious feature must be *correlated*, not merely present. Key the shift of Eq. 6 to a per-emitter random bit drawn independently of the label, at the same magnitude d . Every agent still encodes messages badly, and still does so in a source-dependent way; the only thing removed is the correlation between that dependence and a label shared across agents. If $R_{\mu c}$ stays at zero, the effect is not a consequence of

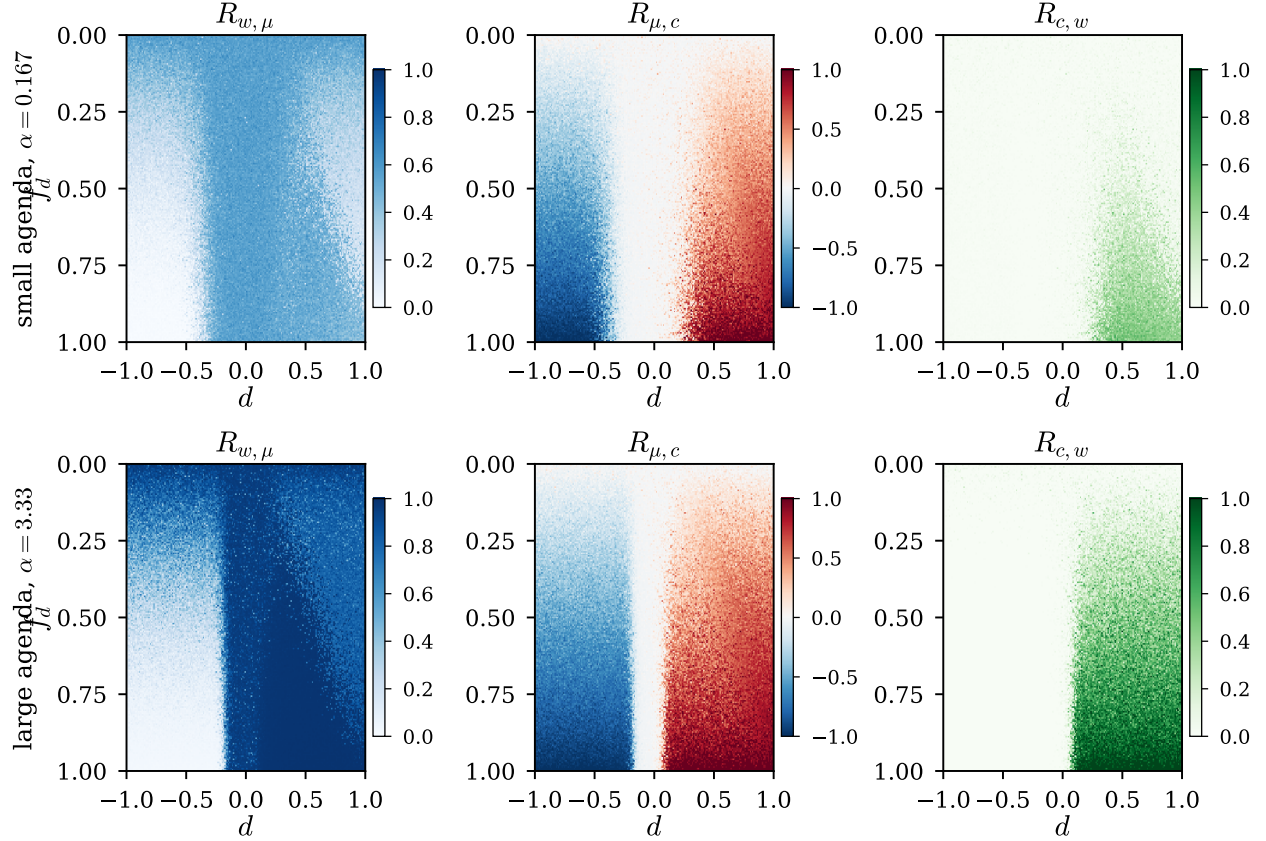


Figure 3: The three correlations of Eq. 8 over the same plane, at low task diversity (top) and high (bottom). The R_{cw} panel is what separates regions (III) and (IV), and its dependence on α previews §8.

	Condition	Prediction	What it establishes
A0	Random-label control	$R_{\mu c} \approx 0$	correlation, not noise
A1	Frozen trust sector	no transition in R_{cw}	trust is the carrier, not the readout
A2	No anti-learning	no polarisation at any d	the sign reversal is the engine
A3	Hebbian / Perceptron	no label-correlated phase	specific to reliability-weighted inference

adding uninformative features, and the paper’s subject is the correlation rather than the noise. This is the control the source draft gestures at in prose and never runs, and it is the cheapest of the four.

A1 — the trust sector carries the split rather than reporting it. Freeze $F_\mu = F_V = 0$, holding every $\mu_{J|I}$ at its initialisation. Note that $R_{\mu c}$ is then trivially near zero and proves nothing — the informative measurement is R_{cw} . If the opinion sector alone still develops a label-correlated split, learned reliability is merely where the split becomes visible; if it does not, reliability estimation is load-bearing. We expect the latter, and the claim in the title depends on it.

A2 — the sharpest test. Replace the prefactor in F_w ,

$$(1 - 2\Phi(h_\mu)) \longrightarrow \max(0, 1 - 2\Phi(h_\mu)), \quad (10)$$

so that a distrusted source produces *no* update rather than a reversed one. Nothing else changes: the same Bayesian machinery, the same trust dynamics, the same perturbation. This removes the effective antiferromagnetic coupling and should remove polarisation entirely, including the label-free polarisation at

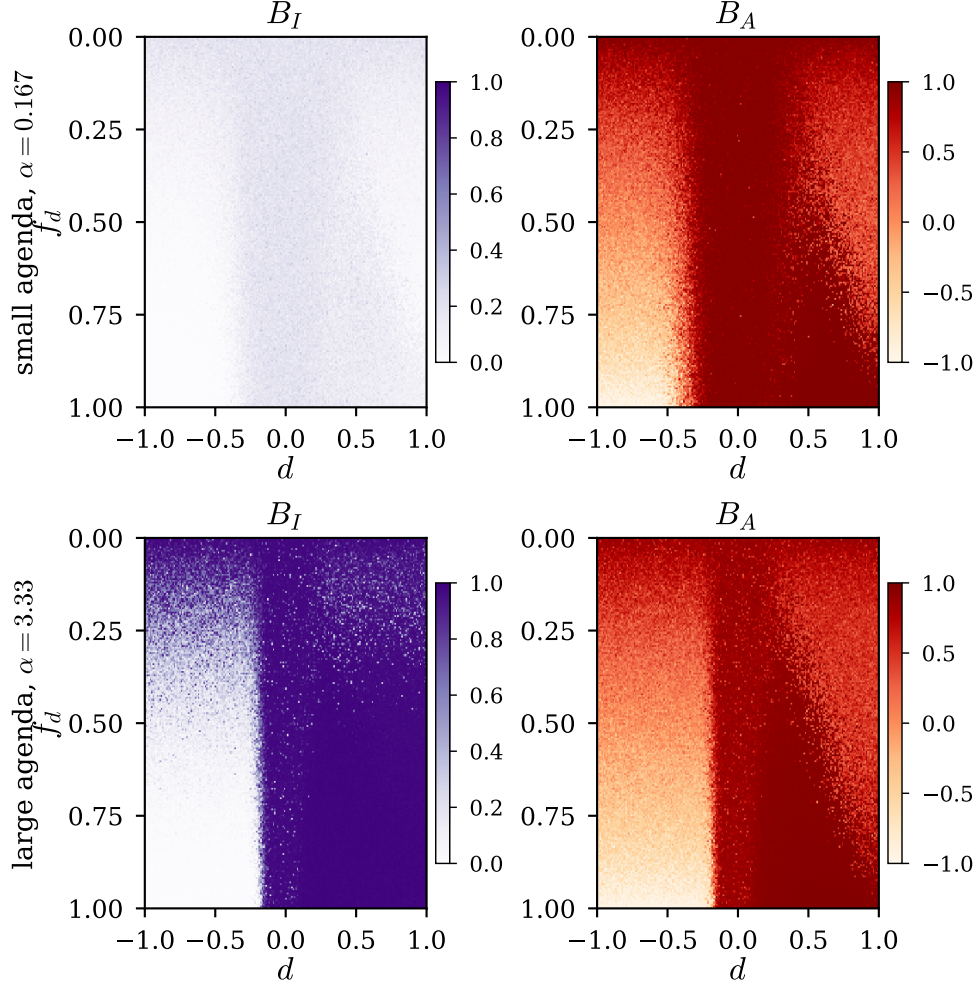


Figure 4: Balance in the opinion sector (B_ρ) and the trust sector (B_η), same plane, two task diversities. [TODO: axis labels still read B_I/B_A ; regenerate.]

$d = 0$. If it does, the mechanism is a single sign in a single prefactor, which is about as sharp as a mechanism claim gets — and it is directly actionable, because a system designer who declines to anti-learn from unreliable peers pays a sample-efficiency cost and buys immunity to this failure.

A3 — it is not learning-from-peers in general. Run the same population, the same perturbation and the same grid under a Hebbian update and under a Perceptron update. The source draft asserts that neither exhibits the collective behaviour; asserted, this is worth little, and shown, it is the result that generalises the paper beyond its own model. It says the phenomenon belongs to rules that maintain and act on a per-source reliability estimate, which is the class of rules the introduction is about.

[TODO: figure: 4 panels $\times$ the (d, f_d) plane, sharing the colour scale of Fig. 2, with the unablated result repeated as a fifth reference panel. One figure, one page of text.]

Reading the four together. A0 says the perturbation must be label-correlated. A1 says the trust sector is necessary. A2 says a specific sign in the trust-to-opinion coupling is necessary. A3 says rules without the trust sector do not do this at all. Together they narrow the cause from “a complicated simulation” to one identified coupling, and they are falsifiable individually.

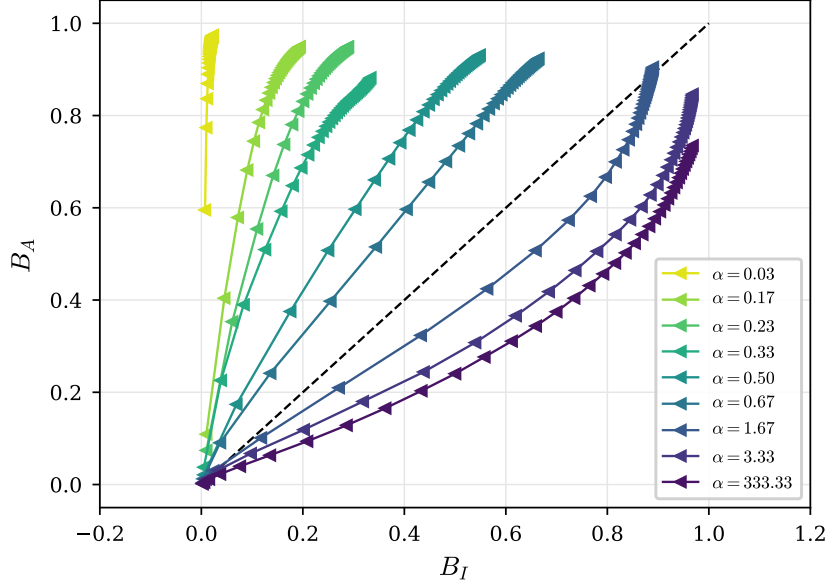


Figure 5: Trajectories in the (B_ρ, B_η) plane for nine task diversities $\alpha = P/K$, from a random initialisation with roughly half of all triples frustrated. Every trajectory ends near the balanced corner: learning anneals frustration away. *Which side of the diagonal it takes to get there is set by α .* [TODO: axis labels still read B_I/B_A ; regenerate.]

8 Task diversity decides what crystallises first

This section is independent of the spurious feature — it holds at $d = 0$ — and is, we think, the most transferable result in the paper.

Start the population from random weights and random trust, so that roughly half of all triples are unbalanced and $B_\rho \approx B_\eta \approx 0$. Learning drives both balance measures toward 1: by moving out of the dissonant quadrants of Eq. 3, agents anneal frustration away. The endpoint is not the interesting part. The *path* is.

The result. For small $\alpha = P/K$ — few issues relative to the representation dimension — the trajectory runs *above* the diagonal: B_η rises first, and the opinion sector follows. For large α the trajectory runs *below* it: opinions organise first and trust follows. The crossing is monotone in α and the reversal happens near $\alpha \sim 1$.

What it means. A population of learners exchanging supervision develops two kinds of structure — agreement about the tasks, and a relational structure over sources — and which of them forms first is not a property of the learning rule but of the task distribution. When the agenda is narrow, there is not enough task signal to organise representations, so the population organises its trust relations first and those relations then pull representations into alignment. When the agenda is broad, representations have enough to lock onto and trust is inferred from the resulting agreement. The symmetry $F_w(x, y) = F_\mu(y, x)$ noted in §3 is why both orders are available to the same dynamics; α selects between them.

Why it matters for the rest of the paper. It explains the difference between the two rows of Fig. 3. At low α and strong d , R_{cw} is small while $R_{\mu c}$ saturates — region (IV) — because the trust structure formed first and never needed the opinion sector to agree with it. The two discriminatory regimes of §6 are therefore not two arbitrary patches of a diagram; they are the same two orders of crystallisation, selected in one case by α and in the other by d .

Practical reading. A system whose agents exchange supervision over a narrow task distribution will form relational structure before it forms agreement, and that structure will be available for a spurious feature to align itself with before there is enough task signal to contradict it. Narrow agendas are the dangerous regime, and they are the common one.

[TODO: state the α values and the measurement time in-text. The source draft leaves Δt as a literal placeholder and it is load-bearing here: because the dynamics anneals rather than equilibrating (§3), a trajectory figure is meaningless without the time at which it was cut.]

9 Related work

[TODO: this section has to be written essentially from scratch. The source draft carries 17 references, all statistical mechanics and 1990s online learning, and the most recent machine-learning-adjacent one is by the same authors. That alone will draw a reject at this venue. The five paragraphs below are the required coverage, with the connection each has to make.]

Shortcut learning and spurious correlations. [TODO: Geirhos et al. on shortcut learning; group-robustness work (Sagawa et al., group DRO); proxy variables in algorithmic fairness. *Connection to make:* the perturbation of §4 is precisely a spurious feature in this sense. Everything in that literature concerns a single model trained on a fixed dataset; the contribution here is that a population of learners exchanging supervision *amplifies* such a feature rather than averaging over it.]

Learning from unreliable sources. [TODO: Dawid–Skene and the crowdsourcing lineage; modern annotator modelling; per-annotator reliability in preference learning and reward modelling. *Connection to make:* the trust sector is a per-source reliability posterior of exactly this kind. That literature asks whether reliability estimation recovers the true labels; we ask what it does to the population of estimators, and find that the same component that confers robustness is the one that transmits bias.]

Decentralised and federated learning. [TODO: gossip and decentralised SGD; client reweighting; Byzantine-robust aggregation. *Connection to make:* this is where per-source weighting actually ships. The distinction worth drawing is that peer weights there are usually a fixed hyperparameter or the output of a robust aggregation rule, whereas here they are a learned posterior with its own dynamics — and it is the dynamics that produces the transition. A robust-aggregation rule with static weights corresponds roughly to ablation A1.]

Feedback loops in populations of learners. [TODO: model collapse and self-consuming training loops; recommender feedback loops; emergent conventions in multi-agent learning. *Connection to make, and it must be explicit:* model collapse is the nearest relative and a reviewer will raise it. The distinction is that there the data distribution itself degrades under recursive training, while here the task distribution is fixed and stationary throughout — the pathology is in how supervision is *weighted*, not in what is available to learn from.]

Statistical mechanics of learning and of opinion dynamics. [TODO: the source draft’s own lineage — online learning in the thermodynamic limit, Bayesian online learning, entropic dynamics — plus structural balance in networks, for the observables of Eq. 9. *Connection to make:* this supplies the tools, and the reason the mechanism is legible at all is that the update rule is derived rather than heuristic. Keep this paragraph short. It is where the paper came from, not what it claims.]

10 Limitations and discussion

Single-layer agents. Each agent is a perceptron. This is a deliberate choice — it is the simplest model in which two agents can agree on some tasks and disagree on others, and it keeps the update rule in closed form, which is what makes the mechanism identifiable at all. It is nonetheless the obvious objection, and §6 carries one experiment outside it. **[TODO: make sure that experiment exists before submission; without it this paragraph is an admission rather than a limitation.]**

The perturbation is injected, not learned. We shift the opinion field by a constant and study the consequences. We do not model how an agent would come to encode messages in a source-dependent way in the first place. That is a separate question and we make no claim about it.

Scale. $N = 40$ agents, and the balance observables cost $O(N^3)$ before the closed form of §5. The finite-size scaling in §6.1 is what licenses reading the boundary as a genuine transition rather than a feature of a small population, and it is the analysis to trust over any single map.

The dynamics anneals rather than equilibrating. $F_C, F_V < 0$ almost everywhere, so the population slows down instead of reaching a stationary state. Every observable is therefore reported at a stated measurement time, and the qualitative structure — which regime, which side of the diagonal — is stable across the range we checked, while the numerical values are not time-independent. We regard time-to-organise as part of the phenomenon rather than a nuisance; §8 is entirely a statement about it.

On the social reading. These are single-layer networks, and extending any of this to human behaviour is not automatic. We use the words group, trust and discrimination because they name the structures the model actually contains — a label carrying no task information, a learned per-source reliability, and a partition aligned with the label — and not as a claim about people. What the result does support is narrower and, we think, more useful: a mechanism by which systems that learn from each other can acquire and entrench a partition along a feature that no component of the system was ever told to care about, without biased data and without a preference anywhere in the system.

Objective misspecification by scale. The update rule is optimal for a teacher–student pair. The two properties that make it good at that job — moving away from sources it has learned to distrust, and attributing surprise to whichever sector is less certain — are exactly the two that produce the collective pathology. This is not a defect in the derivation. It is what happens when an objective derived at one scale is deployed at another, and it suggests that the right question to ask of a peer-weighting scheme is not whether it is optimal for a pair but what it does to a population. Ablation A2 gives a concrete instance: declining to anti-learn from unreliable peers costs sample efficiency and buys immunity to this failure mode.

Reproducibility statement

[TODO: point at the released code, the parameter table (Appendix D), and the seed protocol.]

References

A Derivation of the update rule

Everything the source draft spends its §II on belongs here. The main text needs Eqs. 2–5 and the two properties of §3; a reader who wants to check the derivation wants it complete and in one place.

Order to present it in.

1. The likelihood. Two corruptions, marginalised: the channel flip with probability $\varepsilon(z) = \Phi(z)$, $z \sim \mathcal{N}(\mu_{e|r}, V_{e|r})$, and the encoding mismatch with $\mathbb{E}[x_i y_i] = v_p$, $v_p = \|\mathbf{w}\|^{-2}$. Result: $L(\sigma_e | z, \hat{\mathbf{x}}, \mathbf{w}) = \Phi(z)\Phi(-\mathbf{w} \cdot \hat{\mathbf{x}} \sigma_e) + (1 - \Phi(z))\Phi(\mathbf{w} \cdot \hat{\mathbf{x}} \sigma_e)$.
2. Why $\varepsilon = \Phi(z)$ rather than a Beta posterior. The Beta choice is the natural one and breaks conjugacy; Φ keeps the state in the Gaussian family of Eq. 1. State this as the modelling decision it is.
3. The evidence. Gaussian integrals give Eq. 3. Note the exclusive-or form explicitly — it is the cleanest statement of what surprise means in this model and it is worth a sentence rather than being left to the reader.
4. The maximum-entropy projection back onto Eq. 1, matching first and second moments. This is the step that makes the rule an instance of entropic dynamics and the step that makes it *optimal for a pair*, which the introduction leans on. Give it properly.
5. The modulation functions as log-derivatives of the evidence, Eq. 4, and the resulting update Eq. 5. One presentation, not three. The source draft gives the same update in three notations (its Eqs. 9–12, 13–16, 17–20); collapse them.
6. The symmetry $F_w(x, y) = F_\mu(y, x)$, $F_C(x, y) = F_V(y, x)$, and why the scaling by γ_C, γ_V is what produces it.

Numerical safeguards. The exact modulation functions diverge where $Z \rightarrow 0$, deep in the dissonant corners, and $F_V < 0$ drives V monotonically to zero, so floors on Z and V are needed for a long run to survive. The covariance update $C \rightarrow C + a(C\hat{\mathbf{x}})(C\hat{\mathbf{x}})^\top$ stays positive semi-definite only while $1 + a\hat{\mathbf{x}} \cdot C\hat{\mathbf{x}} \geq 0$, and F_C is unbounded below, so a is clipped at that boundary. Report the floors, report how often the clip triggers, and say plainly that only the clip could alter a trajectory. The source draft mentions none of this; a reviewer who tries to reimplement will hit all three.

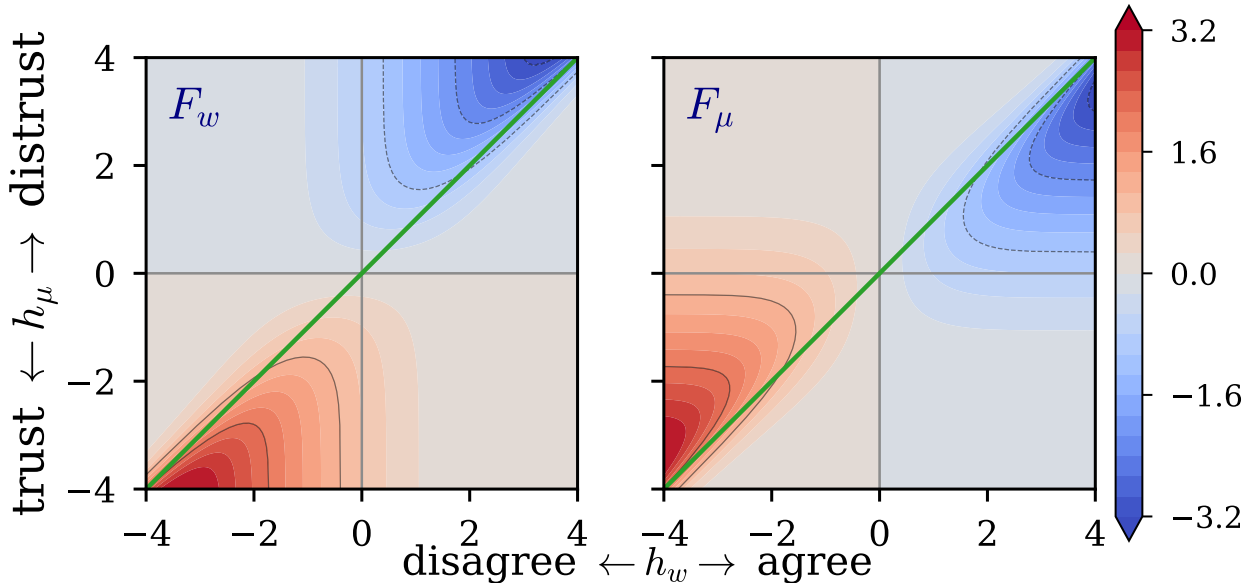


Figure 6: The four modulation functions of Eq. 4 over (h_w, h_μ) . The diagonal $|h_w| = |h_\mu|$ is the blame-attribution crossover of §3; the sign change of F_w across $h_\mu = 0$ is the anti-learning that ablation A2 removes. **[TODO: axes are currently labelled with the unscaled fields; relabel as h_w, h_μ — this is flagged in the source draft too and never fixed.]**

[TODO: modulation_surfaces and modulation_slices are also available. Pick one of the three for the appendix; three views of the same four functions is more than this needs.]

B Ablation specification

Precise definitions for §7. Each condition changes exactly one thing; the grid, the seeds, the initialisation, the measurement time and every other parameter are those of Appendix D.

A0 — random-label control. Draw an independent bit $b_I = \pm 1$ per agent, uncorrelated with κ_I and resampled per seed. Apply Eq. 6 with $D[b_r, b_e]$ in place of $D[\kappa_r, \kappa_e]$, same d , same f_d . Report $R_{\mu c}$ measured against the *true* label κ , and additionally against b — the second is the more informative measurement and tests whether the mechanism aligns structure with *whatever* label the perturbation is keyed to, or specifically with a label that is shared and visible. *Prediction:* $R_{\mu c}[\kappa] \approx 0$ throughout; degraded generalisation relative to $d = 0$.

A1 — frozen trust. Set $F_\mu = F_V = 0$ for all interactions, holding $\mu_{J|I}$ and $V_{J|I}$ at initialisation. The opinion sector updates normally, still reading h_μ from the frozen μ . *Report* R_{cw} , *not* $R_{\mu c}$: with μ frozen at $U(-1, 1)$, $R_{\mu c} \approx 0$ by construction and measures nothing. *Prediction:* no transition in R_{cw} , establishing that the trust sector is load-bearing rather than a readout. Run also with μ frozen at 0 (uniform trust) to separate “frozen at random” from “no trust structure”.

A2 — no anti-learning. Replace the prefactor of F_w by $\max(0, 1 - 2\Phi(h_\mu))$, leaving F_μ, F_C, F_V untouched. A distrusted source now produces no opinion update instead of a reversed one. *Prediction:* polarisation collapses at every d , including $d = 0$; both balance measures stay near zero. If instead polarisation survives, the mechanism is not the sign reversal and the paper’s central claim needs restating — this is the condition most able to falsify it, which is why it is worth running first.

A3 — rule baselines. Same population, same perturbation, same grid, with the opinion update replaced by:

- **Hebbian:** $\hat{w}_r += \frac{\lambda}{\sqrt{K}} \sigma_e \hat{x}$.
- **Perceptron:** the same, applied only when $\text{sign}(\hat{w}_r \cdot \hat{x}) \neq \sigma_e$.

Neither carries a trust sector, so Eq. 6 must be applied to the quantity that plays the role of h_w — i.e. to $\hat{w}_r \cdot \hat{x} \sigma_e$ before the update decision. Say so explicitly; it is the one place where the comparison could be made unfair by accident. Match λ so that the baselines reach comparable generalisation at $d = 0$, and report that calibration. *Prediction:* $R_{\mu c} \approx 0$ throughout for both.

Reporting. One figure, five panels over the (d, f_d) plane sharing the colour scale of Fig. 2: unablated, A0, A1, A2, A3. Seeds and error bars as in Appendix D. Every prediction above should be stated in the text *before* the figure, so that the ablations read as tests rather than as observations.

C Asymmetric perturbations

The main text uses the fully symmetric case, Eq. 7, in which both groups favour their own and are hostile to the other. Five asymmetric variants are implemented, and they are the natural place to ask which structural ingredient the transition actually needs.

D/d	Reading	Question it answers
D_1	$\begin{pmatrix} +1 & 0 \\ 0 & 0 \end{pmatrix}$	one group, in-group tolerance only
D_2	$\begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}$	one group, out-group intolerance only
D_3	$\begin{pmatrix} +1 & -1 \\ 0 & 0 \end{pmatrix}$	one group, both
D_4	$\begin{pmatrix} +1 & 0 \\ 0 & +1 \end{pmatrix}$	both groups, in-group tolerance only
D_5	$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$	both groups, out-group intolerance only
D_6	$\begin{pmatrix} +1 & -1 \\ -1 & +1 \end{pmatrix}$	both groups, both (<i>main text</i>)

Sign convention. Rows index the receiver’s label, columns the emitter’s, and the signs follow §4: a positive entry shifts h_w toward agreement, hence toward trust. So +1 on the diagonal is in-group tolerance. **This is the opposite of the sign printed in the source draft’s Table I**, whose Eq. 25 and table read literally place the discriminatory phases at $d < 0$ while its text and all of its figures place them at $d > 0$. One global sign reconciles the two, and the reading above is the one the sign of F_μ forces. Fix this before the phase diagram becomes a headline figure — the x axis of the main result cannot be ambiguous.

[TODO: report the four correlation maps for D_1 – D_5 . The two rows worth discussing in text are D_1 vs. D_2 (does tolerance or hostility drive it?) and D_3 vs. D_6 (does the effect need both groups perturbed, or does one suffice to organise the whole population?). The second is the more interesting question and neither the source draft nor the main text here answers it.]

D Parameters and protocol

Parameter	Value	Provenance
K , representation dimension	30	recovered from the $\alpha = P/K$ labels of the source’s trajectory figure
N , population size	40	chosen; balance observables cost $O(N^3)$ before the closed form
P , agenda size	5 and 100	chosen, straddling $\alpha = 1$, where the order in §8 reverses
Δt , interactions/channel	500	calibrated against the source’s trajectory endpoints
C_0, V_0	$I, 1$	uninformative Gaussian prior
μ_0	$U(-1, 1)$	starts with $\approx$ half of all triples frustrated
class split	$N/2$ each	
perturbation matrix	D_6 (Eq. 7)	the symmetric case; others in App. C
seeds per grid point	[TODO: ≥ 5]	[NEEDS EXPERIMENT: currently 1]

On $K = 30$. This is recovered rather than chosen. The source draft’s trajectory figure is labelled $\alpha = P/K \in \{0.03, 0.17, 0.23, 0.33, 0.50, 0.67, 1.67, 3.33, 333.33\}$, which are exactly $P/30$ for $P \in \{1, 5, 7, 10, 15, 20, 50, 100, 10^4\}$.

On Δt . The source draft leaves the measurement time as a literal placeholder. Because the dynamics anneals rather than equilibrating (§3), this is not a detail: no trajectory or map in the paper is meaningful without it. State it, state how it was chosen, and show that the qualitative conclusions are stable across a range of it. **[TODO: a small panel sweeping Δt over a decade, showing region assignment is unchanged, forecloses the obvious objection at negligible cost.]**

Seeds. **[NEEDS EXPERIMENT: every map in §6 is currently a single realisation, which is visible in the pixel noise. Repeat at ≥ 5 seeds, report the mean, and put error bars on the extracted boundary of §6.1. This is cheap and its absence is expensive.]**

Deviations from a literal reading of the source draft. Three, each made because the literal reading does not work, and each worth stating rather than silently adopting:

1. **Sign of the perturbation.** See App. C.

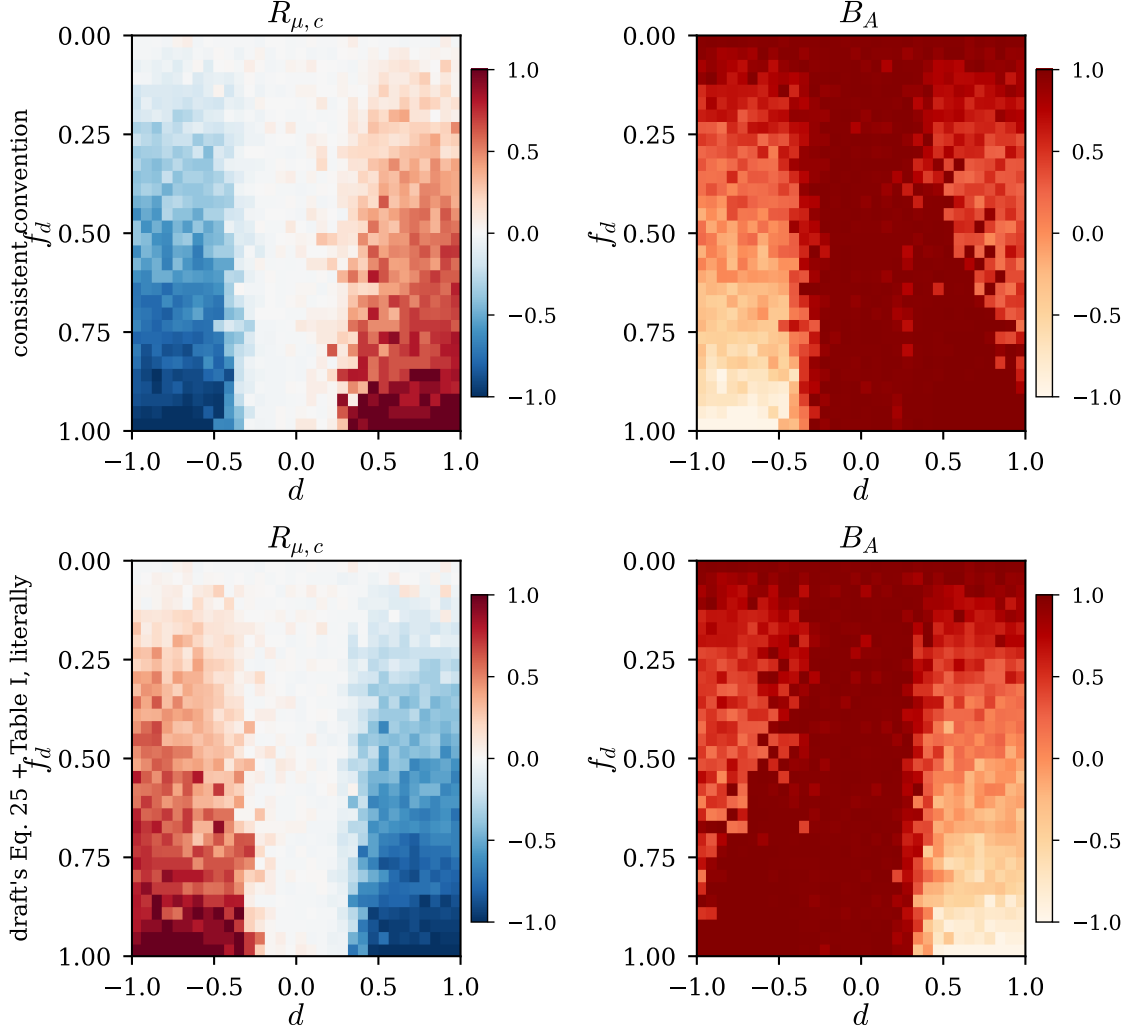


Figure 7: The two readings of the perturbation’s sign, side by side, at otherwise identical parameters. Included so that a reader coming from the source draft can see immediately which convention each figure uses.

2. **The class indicator.** The source defines $G_{IJ} \in \{0, 1\}$, but its own $R_{\mu c}$ map spans $[-1, 1]$, which a $\{0, 1\}$ indicator cannot reach — under perfect reverse discrimination it returns $-(N/2 - 1)/(N - 1)$, and its range depends on N . Since $d < 0$ is half the phase diagram, we use the signed $G_{IJ} = \kappa_I \kappa_J$.
3. **Normalisation of R_{cw} .** The source normalises all three correlations by $N(N - 1)$. That is right for the two that sum $\eta_{I|J} + \eta_{J|I}$ over unordered pairs, but R_{cw} has one term per pair and the same divisor caps it at $1/2$. We use $2/(N(N - 1))$ there.